

K Subset Xors

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers. Consider the xors of all 2^n subsets of the array (including the empty subset with xor equal to zero).

Your task is to find the k smallest subset xors.

Input

The first line has two integers n and k : the size of the array and the number of subset xors k .

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print k integers: the k smallest subset xors in increasing order.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq k \leq \min(2^n, 2 \cdot 10^5)$
- $0 \leq x_i \leq 10^9$

Example

Input	Output
4 9 3 5 14 8	0 0 3 3 5 5 6 6 8